

User Need and AI Contribution

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Retrieval Augmented Generation

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Numbers and Dates

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Multimodal Evidence

Without opening the old lesson, explain one key idea from Multimodal Evidence. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Capstone requirements and corpus. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone requirements and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-evidence test.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

6. Solve the given case

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

Narrow the need and AI contribution.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A student needs the exact due date from a course notice. Should the design start by generating a plausible date or locating the notice? Explain the AI contribution.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 19 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: A complete capstone plan is not evidence that the implementation meets it; evaluate after building. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-evidence test.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: User Need and AI Contribution

Define a useful reading helper before choosing its AI features.

A pupil needs the meaning of unfamiliar words in an approved glossary. G3 defines habitat as the place an organism lives; no other sources are allowed.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

User Need and AI Contribution

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the need

The pupil wants a glossary meaning, not a made-up example presented as a fact.



2. Choose contribution

The assistant locates the glossary entry and restates it in clear language with G3.



3. Bound the service

An absent word produces "not in this glossary"; a direct glossary search is the baseline.

CHANGE ONE THING

The pupil asks for photosynthesis, which G3 does not contain.

Explain the changed result using the matching lesson.

What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Define a useful reading helper before choosing its AI features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? An absent word produces “not in this glossary”; a direct glossary search is the baseline.

1. Bound the service
2. Name the need
3. Choose contribution

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bound the service An absent word produces “not in this glossary”; a direct glossary search is the baseline. Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The pupil asks for photosynthesis, which G3 does not contain. Which conclusion is supported by the supplied facts?

1. The assistant reports that the definition is unavailable in its approved glossary.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The assistant reports that the definition is unavailable in its approved glossary. A helpful interface cannot expand its authorised evidence by guessing.

Repair a missing building block

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the mechanism and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Which part can be corrected immediately from existing evidence, and which needs follow-up?. Return to this week and explain what you changed.

Review Multimodal Evidence (ai-3/13). Goal: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
